

draws. If one criticism were to be made of an otherwise fine volume, it might be that the repeated focus on specific directors occasionally venerates the auteur in ways that the book's premise cautions against. These criticisms, however, are equally balanced by the necessity of engaging cinema's great architects. The volume is especially helpful in highlighting the work of film artists and theologians of whom many in the English-speaking world might be unaware. The book's interdisciplinary roots offer a fine reminder that theology is not a tool awaiting application to a cinematic artefact nor is it a prescribed method to be replicated by dolt. Rather, cinema speaks in its own voice (to once more confuse a metaphor) and a theological appreciation of the medium is one characterised by an attentiveness to new meanings as they emerge from sites of culture that may not necessarily be seen as religious, theological, or even Christian alongside an appreciation of other voices across the humanities and cultural studies with which theological reflection finds affinities and sites of encounter.

The Digitalised Image of God: Artificial Intelligence, Liturgy, and Ethics. By

Ximian Xu. London and New York: Routledge, 2024. Pp. x + 210. Price £105.00. ISBN 9781032397832.

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It hardly needs to be said that Artificial Intelligence has become ubiquitous and that it impinges on our lives in myriad ways, from the apps on our smartphones to the large-language models (LLMs) which power services like ChatGPT. Many people will also be aware of the philosophical and ethical issues to which the burgeoning power of AI gives rise. Could a machine ever be said to think? Could such a machine ever be said to be conscious? What ethical status ought such a machine to have? Present-day AI simply crunches data, producing responses to particular tasks from the vast amount of material to which it has access (hence the issues raised by the pirating of copyrighted texts in the training of LLMs). But looking to a possibly not-too-distant future, what might we say about Artificial General Intelligence (AGI), an hypothesised future achievement at which point a machine's intelligence might be said to equal that of a human? And what about Artificial Superintelligence (ASI), at which point that intelligence exceeds that of humans—leading to eschatological visions such as the “Singularity,” at which point humans might find themselves redundant?

A considerable literature already addresses such questions as these; but explicitly theological responses to them remain thin on the ground. All the more reason, then, to welcome Ximian Xu's “The Digitalised Image of God” for the insights which it brings to current discussions. An introduction locates Xu's work in the contemporary field of AI studies, and two chapters then outline his theological approach. As might be expected from his title, Xu draws on the concept of the *imago Dei*, noting that this should be seen in holistic terms, encompassing humans as embodied (not simply “spiritual”) creatures. Drawing on the work of Noreen Herzfeld and others, he suggests that AI might be seen as being *imago hominis*—bearing the image of humans, who in turn bear the image of

God. In a novel twist, Xu then introduces the notion of archetype and ectype, as expressed in the Reformed theology of writers such as Herman Bavinck. The idea here is that archetypal concepts relate to uncreated reality (i.e. God), and ectypal concepts to created realities: this creates an “ontological chasm” (p. 32) between the two. Xu summarises Bavinck’s approach thus: “*the whole human being ... is the imago Dei and the ectype of the divine archetype*” (p. 34, italics in original). Xu then maintains that “as the human being is the ectype of God who is the *uncreated* archetype, AI is the ectype of the human being who is the *created* archetype” (p. 41). He proceeds to discuss the nature of any consciousness that might be attributed to an AI, arguing that “artificial consciousness is the ectype of human consciousness and is attributed [by humans] with religious and moral characteristic features (e.g., intentionality)” (p. 68).

In subsequent chapters, Xu employs this theological lens as a means of analysing in depth three important issues, which are already with us: the use of AI in liturgical contexts, the moral status of AIs, and the use of AIs as companions for human beings. In these cases, rather than accept understandings of AI which rule that it does or does not possess an equivalent status to human liturgical leaders, moral agents, or personal companions, he offers the ectypal understanding of AI which he has developed as a pointer towards a more nuanced way forward.

With regard to worship, Xu notes the potential of AI to extend the reach of conventional Church services, through replicating them when a human is not present and through developing online versions (although he cautions that this can lead to an over-individualisation of the worship experience). He observes that an enactment of a liturgy by a human cleric is conscious, and involves memories (those of the cleric and the congregation), whilst such enactment by a machine simply involves the computation of data stored by that machine (p. 97), thereby limiting the extent to which an AI might fully replace a human cleric. He states, “Although the idea of AI as the ectype of humanity brings forth the close connection between clerics and AI ministers, the ectypal implication presupposes the ontological distinction between the archetype and the ectype. ... AI-powered liturgical enactment should give way to the human enactment of liturgy in the event that the latter is viable” (pp. 109–10).

When considering whether or not machines can be “Artificial Moral Agents (AMAs)”, Xu notes that responsibilities for their behaviour must always, ultimately, rest with those who have created them (even if they have been designed to learn “by themselves”). He points out that “The HMA [Human Moral Agent] cannot escape moral responsibility after deploying computational artefacts to make ethical decisions. It is always the HMA who designs and uses computational artefacts to address ethical dilemmas” (p. 135). This leads Xu to develop the idea of “partial moral agency”: noting that “the AMA only performs certain moral actions to address particular moral issues according to the goals established by AI researchers” (p. 140), he states that “*partial* artificial moral agency refers to the integration of responsible AI within human moral life made by responsible HMAs through human ectypal creation of AI in the *imago hominis*” (p. 148, italics in original).

It is perhaps in Xu’s final chapter, exploring that idea of AI “artificial companions,” that his theological critique is most effective. Xu explores the crucial part played by empathy in the development of relationships, noting that this is again a “whole-body”

phenomenon; and he discusses the various proposals regarding artificial companions which have been made thus far, from “carebots” intended to provide companionship for the elderly and housebound (which are already with us) to the creation of robot lovers (which, thankfully, are not). Xu argues persuasively for the use of the term “companionable AI” in preference to “artificial companion,” pointing out that this “shift[s] the focus from ‘companion’ to AI itself, implying the ontological distinction between artificial and human companions” (p. 175); and he urges that “AI as the ectype of humanity provides a solid onto-theological basis foundation for companionable AI” (p. 182), since it reinforces that distinction.

Scattered throughout the book are references to initiatives which are already being undertaken within Church and other circles to respond to issues raised by AI. These include the recent report published by the Church of Scotland (p. 40) and the “Edinburgh Declaration on Responsibility for Responsible AI” (p. 146). Xu also cites “case studies” of AI in action, such as a 2023 event at which a Lutheran service in Nuremberg was taken by Chat GPT, producing ambivalent responses in the congregation (pp. 77–78). Given the full-on rush to develop AI at the present time, these are all valuable and important instances of ways in which consideration is being given to the questions which it raises.

Xu’s book is firmly anchored in contemporary theological scholarship and extends that scholarship in valuable ways. It is a dense read, and Xu’s language is sometimes convoluted. I was not always fully convinced by his ectypal approach: though persuasive in the cases in which it is employed here, I wonder how effective it will prove as AI develops further, particularly if it should become possible to ascribe personal histories to AIs, or should AGI or ASI ever be achieved. However, there is much that is important here in deflating some of the language around AIs, in drawing attention to the responsibilities of those who programme them, and in setting out some theological parameters within which to engage issues concerning AI—which is raising some of the most pressing questions of our time.